

MUHAMMAD SYAHMI BIN ABDUL RAHMAN

Engineering Project Manager | Robotics, Autonomous Systems & Industrial IoT

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PROFESSIONAL SUMMARY

Engineering project manager with 7+ years delivering robotics, autonomous systems, and Industrial IoT (IIoT) platforms from R&D through production deployment. Builds autonomous robots, the operator GUIs that drive them, and the IIoT platforms that monitor them, from concept to commissioning. Currently leads a cross-functional team of 6 to 7 engineers across software, electrical, and mechanical disciplines; architected the ROS2 to MQTT communication backbone, shipped the in-house FleetOps real-time fleet management platform, and delivered an RFID and smart-weighing clinical-waste tracking system on site. Improved engineering delivery efficiency by 20%, increased client retention by 25%, and sustains 75% fleet uptime. Targeting Engineering Manager roles focused on scalable robotics architecture, team development, and product-driven automation strategy.

CORE COMPETENCIES

Engineering Team Leadership, Robotics & Autonomous Systems, Industrial IoT (IIoT) Integration, System Architecture & Design, ROS2 & MQTT Frameworks, Operator GUI / HMI Design, SLAM & Mapping Tools, RFID & Asset Tracking, Fleet Management Platforms, Agile & Sprint Planning, Product Development Lifecycle, Cross-Functional Team Management, Stakeholder Communication, Risk Management & Mitigation, Budget & Resource Planning, PLC Programming, Edge & Embedded Systems, Technical Documentation & SOPs.

PROFESSIONAL EXPERIENCE

Project Manager, R&D and System Integration

Sailcott (M) International Sdn. Bhd.

2020 - Present

Shah Alam, Malaysia

Lead a cross-functional team of 6 to 7 engineers across software, electrical, and mechanical disciplines, driving end-to-end development and deployment of robotics and automation systems from R&D through commercial commissioning.

- Directed development of autonomous mobility platforms (Electric Cart Carrier (ECC) and Autonomous Buggy / UGV) from concept through field trials to production commissioning.
- Shipped FleetOps, an in-house real-time fleet management web platform with live dashboards, geolocation map, multi-camera feeds, battery and speed analytics, role-based access, and time-series telemetry from each robot.
- Delivered an RFID and smart-weighing clinical-waste tracking system on site: UHF tag reads at the weighing station, live bin / gross / tare / net weight on a wall-mounted operator GUI, and automated compliance logging that replaced manual paperwork.
- Designed operator-facing GUIs / HMIs used daily on site (multi-map ROS2 launch, SLAM mapping and teleop, keepout-mask generator, and PLC ON/OFF control over USB-UART), turning a terminal-only workflow into a one-screen operator panel.
- Architected the ROS2 to MQTT communication framework (TLS, QoS, auto-reconnect, configurable topic mapping) that powers ECC, UGV, and FleetOps, surviving flaky cellular links during outdoor field trials.
- Delivered 3 major automation projects ahead of schedule while maintaining full budget compliance, sustaining 75% fleet uptime in field deployment.
- Improved engineering delivery efficiency by 20% through structured sprint planning, workflow optimization, and standardized reporting.
- Reduced integration issues through structured validation, testing, and commissioning protocols.
- Serve as primary technical liaison between engineering teams, management, and clients to align product requirements with deployment timelines.

Automation & IoT Consultant (Freelance, alongside primary role)

Independent Practice

2018 - Present

Shah Alam, Malaysia

Provide industrial automation, PLC programming, and IoT system integration services to SMEs across manufacturing and energy sectors.

- Delivered 15+ automation and IoT projects including PLC systems, CRM integration, and cloud-based real-time monitoring dashboards.
- Designed MQTT-based IoT architectures and Node-RED dashboards for operational analytics and performance tracking.
- Increased client retention by 25% through proactive system optimization and structured reporting.
- Reduced manual reporting workload by implementing automated monitoring and control systems.
- Owned project scope definition, milestone planning, vendor coordination, and deployment execution end to end.

Automation Project Intern

2018

East Automation & Engineering Sdn. Bhd.

Subang Jaya, Malaysia

- Supported system design, prototype deployment, and validation for solar monitoring and laser surveillance automation systems.
- Developed GUI control interfaces for automated monitoring solutions.

SELECTED PROJECTS

Full case studies, demos, and screenshots available at al-zuji.github.io/resume.

Electric Cart Carrier (ECC)

ROS2, MQTT, NVIDIA Jetson, Python

Autonomous electric cart carrier for clinical-waste handling, delivered concept-to-commissioning. Perception, navigation, and fleet control integrated with the ROS2 to MQTT backbone.

Autonomous Buggy / UGV

ROS2, Embedded Linux, C++, Sensor Fusion

Unmanned ground vehicle delivered ahead of schedule. Cross-disciplinary integration across software, electrical, and mechanical teams, from lab build to outdoor field trials.

FleetOps - Real-Time Fleet Management Dashboard

ROS2, MQTT, WebSockets, React, Node.js, Time-series DB, Mapbox, RTSP/WebRTC

In-house web platform monitoring and managing the autonomous buggy / ECC fleet end-to-end: live status, geolocation map, multi-camera feeds, battery and speed analytics, and role-based admin streaming telemetry from each robot in real time.

RFID Clinical Waste Tracking & Weighing Station

UHF RFID, Smart Scale, MQTT, Python, Node-RED, Edge Gateway, Operator GUI

End-to-end RFID and smart-weighing system deployed on site. UHF-tagged bins auto-identified at the weighing station; bin ID, gross, tare, and net waste weight stream live to a custom operator dashboard with auto-tare logic and automated compliance logging.

Scrubber Robot - Operator Control Panel

Python, PyQt, ROS2 Launch, SLAM Toolbox, PLC / Modbus, FT232R USB-UART

Purpose-built desktop GUI letting site operators run the autonomous scrubber without a terminal. One panel for multi-map ROS2 launch, SLAM mapping and teleop, keepout-mask generation, and direct PLC control, with live battery and connection health.

ROS2 to MQTT Bridge

ROS2, MQTT, TLS, QoS 0/1/2, Auto-reconnect, Docker

Communication backbone behind ECC, UGV, and FleetOps. Bidirectional bridge translating ROS2 topics to MQTT in real time with QoS, auto-reconnect, and configurable topic-mapping. Scales across multiple robot fleets without DDS leaking onto the wider network.

TECHNICAL SKILLS

Robotics & Integration: ROS2, MQTT, Node-RED, REST API Integration, Industrial IoT Platforms, UHF RFID, Fleet Management Systems

Programming & Control: Python, C, C++, PyQt, PLC Programming (Ladder Logic), HMI / GUI Development

Hardware & Edge Systems: NVIDIA Jetson Orin, Raspberry Pi, Embedded Linux, Microcontrollers, FT232R USB-UART, Modbus

Robotics Tooling: SLAM Toolbox, ROS2 Launch, Sensor Fusion, Teleoperation, Map / Keepout-mask Generation

Project & Collaboration: Microsoft Project, Jira, Trello, Asana, Git, Docker

Documentation & Visualization: Microsoft Office, Visio, Draw.io

Methodologies: Agile, Scrum, Sprint Planning, Risk Management, System Validation & Testing

EDUCATION

Bachelor of Engineering, Electronics Engineering

Jan 2017 - Jan 2019

Universiti Teknologi MARA (UiTM), Shah Alam

Diploma in Electrical Engineering (Power)

Jan 2014 - Sep 2016

Universiti Teknologi MARA (UiTM), Dungun

ADDITIONAL INFORMATION

Languages: Bahasa Malaysia (Native), English (Professional Proficiency).

Portfolio: al-zuji.github.io/resume.

References: Available upon request.